Instructor Guide to ‘Science Around the Board’

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March 2026

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Introduction

Educational practices are constantly evolving, moving away from passive absorption towards active engagement. Growing up in a memorization-heavy system, I experienced firsthand the limitations of rote learning and the powerful impact that active, student-centered strategies can have on both motivation and long-term retention. As instructors, our primary goal is to support such deeper learning, but we should not view this as a strictly rigid or purely serious process.

This is where the potential of playful pedagogy becomes transformative. By integrating game mechanics such as competition, resource management, and immediate feedback into the curriculum, we can create a dynamic environment where students feel safe to fail and eager to try again. The game I developed, Science Around the Board, leverages these principles to turn the reviewing into an engaging learning cycle. This manual will guide you through that transition, showing you how to align the dynamic energy of a board game with the rigorous learning objectives of your course.

Definitions

What is *playful pedagogy*?

Playful pedagogy is an instructional approach that integrates the structures and spirit of play, curiosity, experimentation, and social interaction, into the learning process. In this model, teachers transform into facilitators rather than remaining the center of the classroom. This shift establishes a "safe space" where students can take risks, make mistakes, and learn iteratively without the immediate anxiety of grade penalties.

What is *Science Around the Board?*

Science Around the Board is a customizable, web-based, open-source educational platform developed to gamify the learning of complex topics. Instructors provide a question file in tab-separated values (.tsv) format, and optional images for the quizzes, to teams of 1-4 students who independently upload them to the game on their computer. Prior to starting, each student must complete a pre-game survey (pre-assessment), after which players alternate to roll a pair of dice and move along a square board. Each side contains 9 tiles dedicated to a specific theme. Players that fall on an unowned tile must answer an instructor-generated question for the option to buy the tile, while players who fall on an owned tile must pay rent. Players gain chaos tokens by successfully completing milestone quizzes in the corner tiles. Chaos tokens

can be redeemed for a chance at stealing other player's properties. Players who own all tiles of a specific color can add stars to charge exponentially higher rent. The game ends when all but one player go bankrupt or when the class is over. Following the gameplay, students are prompted to complete a post-game survey (post-assessment), enabling instructors to track learning.

The game leverages key pedagogical frameworks, such as the BOPPPS model of lesson design, along with active learning techniques and frequent low-stakes assessments to incentivize and scaffold learning. Instructors should include explanations with each question to provide frequent learning opportunities.

While originally designed for a 4th-year undergraduate research course in Microbiology, the modular nature of the game engine works seamlessly with absolutely any topic, as the content is dynamically determined by the input file. Its versatility is limitless: I have used the exact same engine to run advanced genomics workshops and to host a *RuPaul's Drag Race* trivia night with my wife!

Why use 'Science Around the Board'?

Successfully leveraging play in the classroom requires aligning activities with the learning objectives. Misalignment, requiring effort allocation towards tasks that do not move students forward in the learning process, can be counterproductive by decreasing motivation and attention. It is critical to be intentional, ensuring games align with Learning Outcomes (LOs), and that the fun does not distract from the learning. A useful strategy when brainstorming playful frameworks is to begin thinking about which verb to address in Bloom's Taxonomy.

In my experience as an instructor, TA, and facilitator I recognised that students often struggle to conceptualise complex class material, frequently performing tasks without understanding the theory behind it. Students frequently "do" tasks without "knowing", thus the playful activity I had envisioned had to address the Bloom's verb "understand", which involves gaining more knowledge through revision and practice. *Science Around the Board* was built to bridge this gap: it enables students to step back from the rote execution of tasks and confront the conceptual framework that supports them.

Science Around the Board is best suited as a replacement for revision sessions, with successful execution and student feedback for both individual or group use and synchronously or asynchronously. Instructors can design questions spanning any range of topics and leverage matching pre- and post-assessments to track students' learning throughout the game. Additionally, *Science Around the Board* incorporates numerous opportunities for iterative assessments followed by scaffolded explanations, enabling learning opportunities at every stage while removing the fear of failure.

Important links

Game: https://hghezzi.github.io/Science-Around-the-Board/
Encryption tool: https://hghezzi.github.io/Science-Around-the-Board/encryptor.html
GitHub repo: https://github.com/hghezzi/Science-Around-the-Board

How to use 'Science Around the Board'

Choosing a topic

This is probably the most straightforward portion of the guide. While *Science Around the board* was designed for a microbiology course, instructors can use the game engine for absolutely any topic. Pick a subject that is suitable for your course and that aligns with your specific learning outcomes.

Designing the input file

Science Around the Board requires students to input a file with all questions (required) and any images that may be referenced by the questions (optional). The input file must be in Tab-Separated Values (TSV) format, which I recommend creating with Google Sheets, then exporting as `.tsv`. This question file must follow the header format specified below:

Column Header	Required?	Description	Examples
id	Yes	Unique ID for the row.	bio_01
question	Yes	The text prompt.	What is the start codon?
option1	Yes	Answer Choice A.	AUG
option2	Yes	Answer Choice B.	UAA
option3	Yes	Answer Choice C.	GGC
option4	Yes	Answer Choice D.	AAA
correctIndex	Yes	The numeric position of the correct answer.	1 (for Option 1)

explanation	Yes	Feedback shown after answering.	AUG codes for Methionine.
type	Yes	Controls game logic (see below).	Property, etc…
bigTopic	Yes	Overarching topic to be taught	16S rRNA sequencing
module	Yes	Sub-topic in ‘bigTopic’	Quality control
theme	No	1 of 4 unique themes in each ‘module’	Translation
subtheme	No	1 of 2 unique sub-themes in each ‘theme’	Genetics
imageFile	No	Filename for images, must match filename in images uploaded	diagram_a.png

Here is a screenshot of what the exported questions.tsv file would look like:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	**id**	**question**	**option1**	**option2**	**option3**	**option4**	**correctIndex**	**explanation**	**bigTopic**	**module**	**theme**	**subtheme**	**type**	**imageFile**	
2	prop_seq_me	What is the c	Sequencing,	Sample Prep,	Sample Prep,	Cluster Gene	2	The workflow	16S	QIIME2	Amplicon sec	Sequencing	property		
3	prop_seq_me	During 'Samp	Polymerase e	Fluorescent p	Oligonucleot	Sequencing p	3	Adapters are	16S	QIIME2	Amplicon sec	Sequencing	property	illumina_bridge.png	
4	prop_seq_me	Which specif	Bridge Amplif	Sanger Seque	Emulsion PCI	Shotgun Clor	1	Bridge amplif	16S	QIIME2	Amplicon sec	Sequencing	property		
5	prop_seq_me	What critical	They contain	They are perr	They are mag	They are radi	1	The reversibl	16S	QIIME2	Amplicon sec	Sequencing	property		
6	prop_seq_me	What factor c	The physical	The total amo	The number c	The length of	3	Read length i	16S	QIIME2	Amplicon sec	Sequencing	property		
7	prop_seq_me	How does 'Pa	It sequences	It sequences	It sequences	It sequences	4	Paired-end se	16S	QIIME2	Amplicon sec	Sequencing	property		
8	prop_seq_me	What is the p	To identify wh	To align the s	To measure t	To determine	1	Indices act a	16S	QIIME2	Amplicon sec	Sequencing	property	illumina_bridge.png	
9	prop_seq_me	On what phys	A silicon micr	A glass flow c	A plastic mic	A magnetic b	2	The reaction	16S	QIIME2	Amplicon sec	Sequencing	property		
10	prop_seq_me	How does the	By measuring	By detecting	By measuring	By measuring	2	Each of the fo	16S	QIIME2	Amplicon sec	Sequencing	property	illumina_calling.png	
11	prop_seq_me	What constit	A group of ide	A group of diff	A single DNA	A group of sec	1	A cluster is a	16S	QIIME2	Amplicon sec	Sequencing	property		
12	mile_seq_me	You need to s	Paired-end 1	Single-end 30	Paired-end 3	Single-end 40	3	To cover 460b	16S	QIIME2	Amplicon sec	Sequencing	milestone		
13	mile_seq_me	If you use a 2	The sequenci	The sequenc	The reads are	The sequenc	4	The sequenc	16S	QIIME2	Amplicon sec	Sequencing	milestone		
14	mile_seq_me	Why is 'Phasi	It prevents th	It causes the	It reduces the	It creates a m	4	When strand	16S	QIIME2	Amplicon sec	Sequencing	milestone		
15	mile_seq_me	Why do we ac	To improve th	To increase t	To allow com	To physically	3	Pooling lowe	16S	QIIME2	Amplicon sec	Sequencing	milestone		
16	mile_seq_me	Why is the do	To allow hybr	To wash awa	To visualize t	To stop the ar	1	Double-stran	16S	QIIME2	Amplicon sec	Sequencing	milestone		
17	mile_seq_me	If a nucleotid	Phasing (stra	Pre-phasing (	Deletion (bas	Substitution	1	If the termina	16S	QIIME2	Amplicon sec	Sequencing	milestone		
18	mile_seq_me	Why is the or	To prevent in	To save sequ	To degrade th	To make roon	1	Read 2 requir	16S	QIIME2	Amplicon sec	Sequencing	milestone		
19	mile_seq_me	Why is Paired	It merges rea	It is significar	It produces tw	It is faster to	1	Paired-end se	16S	QIIME2	Amplicon sec	Sequencing	milestone		
20	mile_seq_me	What happer	The quality sc	The clusters v	Demultiplexi	The run proce	3	The index rea	16S	QIIME2	Amplicon sec	Sequencing	milestone		

While the table above provides a quick reference, understanding how these columns interact is the key to mastering *Science Around the Board*. The game engine reads your TSV file and dynamically constructs the board, the menus, and the pre/post surveys based entirely on the text you provide.

Here is a detailed breakdown of how to use each column effectively:

Board hierarchy and organization

These columns act as the blueprint for your game. They tell the engine where to place questions and how to organize the main menu.

- **bigTopic and module:** These define where your game lives in the main menu. For example, if your bigTopic is "Microbiology" and your module is "16S Sequencing," students will select those options to start.
    - *Pro-Tip:* You can use comma-separated lists here! If a question applies to multiple modules (e.g., "16S Sequencing, Whole Genome Sequencing"), the engine will pull that question into both game modes automatically.
- **theme** (the 4 sides)**:** The engine uses the first four unique themes it finds to build the four sides of the game board. Think of a theme as a major chapter of your module.
- **subtheme** (The property groups)**:** Within each side (theme), the engine creates two distinct property groups based on the subtheme column. These are the specific topics students will purchase and defend (e.g., "DNA Extraction" and "Library Prep").

Game logic and rules

- **type:** This is the most critical control column. It dictates *when* and *how* a row is used in the game. You must use specific keywords here:
    - property: Standard questions used when a student tries to buy or defend a regular board tile.
    - milestone: Harder, comprehensive questions used for the "Corner Exams" (students must answer 5 out of 6 of these correctly to capture a milestone).
    - core: Questions dedicated to the mid-board utility tiles (e.g., the Sequencing Core).
    - mishap: Creates random "chance" events (e.g., "Someone left the freezer open! Lose $100."). Put the event text in the question column and the penalty/reward in the explanation column.
    - survey / confidence: These rows bypass the board entirely and are compiled into the mandatory pre-game and post-game assessments to measure learning outcomes. 10 questions among the “survey” options are randomly drawn to display as pre- and post-game surveys to each student. Confidence questions appear as sliders and are the same for all students.

The assessment content

This is the actual material your students will interact with during play.

- **id:** A unique identifier for your own tracking (e.g., WGS_001). This makes it much easier to match post-game data exports to specific learning objectives.
- **question:** Keep this clear and concise. If the prompt is too long, students may experience reading fatigue during the fast-paced competitive gameplay.
- **option1 – option4:** The multiple-choice answers. The engine will automatically randomize the order of these options every time the question appears, so you do not need to worry about randomizing them yourself.

- **correctIndex:** The numeric position (1, 2, 3, or 4) of the correct answer based on your columns.
- **explanation:** This is your pedagogical safety net. This text appears immediately after a student answers, regardless of whether they were right or wrong. Use this space to clarify misconceptions, provide a fun fact, or reinforce the learning objective.
- **imageFile:** If a question requires a visual aid (like a graph, gel electrophoresis result, or code snippet), put the exact filename here (e.g., gel_lane_1.png). *Note: The filename must perfectly match the file you upload or include in your image folder that you share with students.*

For an example input file containing all columns formatted correctly, please visit: *github.com/hghezzi/Science-Around-the-Board* and look for the file '**SAB_questions_Jan22_Filtered.tsv**' inside the '*public*' folder.

Optional: encrypting your questions TSV file

Since the question (.tsv) file you distribute to students **also contains the answers**, I have created an encryptor tool. Simply upload your .tsv question file, choose a **password to share with students**, and you will **receive an encrypted .lock file**. This is the file to share with students together with the associated password.

Game mechanics

Once the file is loaded and the game begins, *Science Around the Board* operates on a blend of resource management, academic trivia, and strategic risk-taking. Here is how the game actually plays out.

The objective

Teams compete against each other to achieve one of two victory conditions:

1. **Last standing:** be the last player not to go bankrupt
2. **Asset dominance:** If time runs out, the player with the highest net worth (cash + asset value) is declared the winner.

The core turn sequence

Each team's turn follows a strict, fast-paced loop:

1. **Roll:** The active team rolls the digital dice.
2. **Move:** Their token advances around the board automatically.
3. **Encounter:** The team interacts with the tile they land on.

Tile encounters

The heart of the game lies in what happens when a player lands on a specific tile.

- **Unowned tiles (properties):** If a tile is unowned, the player may attempt to buy it. They are presented with a multiple-choice question.
    - *Correct answer:* They have the option to pay the tile's cost and take ownership.
    - *Incorrect answer:* They are penalized with a minor cash fine, and the tile remains unowned.
- **Rival tiles (rent defense):** If a player lands on a tile owned by a rival, they owe rent. This triggers the **rent defense** mechanic. The trespassing player must answer a question.
    - *Correct answer:* The rent penalty is reduced by 50%.
    - *Incorrect answer:* The player must pay the full.
- **Milestones (corner exams):** Capturing a corner requires more than just money; it requires mastery. Players must pay an entrance fee and demonstrate mastery in the theme of that specific side of the board, achieved by answering 5 out of 6 questions correctly.
- **Lab mishaps (chance):** These are random, unpredictable events drawn from your TSV file that either reward teams (e.g., "Grant Renewal Approved! +$200") or penalize them (e.g., "Contamination! -$100").

Advanced mechanics: lab management and chaos

To keep engagement high in the later stages of the session, the engine includes two advanced mechanics:

- **Upgrading infrastructure:** If a player manages to purchase all the tiles within a specific `subtheme` (e.g., owning all the "Primer Selection" tiles), they unlock the **Lab Manager**. This allows them to spend cash to upgrade the "level" of those tiles, drastically increasing the rent charged to rivals who land there.
- **Chaos tokens:** When a player successfully captures a Milestone, they are awarded a Chaos Token. These highly coveted items can be spent to launch a "Chaos Challenge" against a rival's property, allowing the attacking player to steal the tile if they can correctly answer a question. Once all 4 milestones have been acquired, chaos tokens can be purchased for $500.

Game surveys and feedback

Science Around the Board integrates a built-in strategy to collect feedback on student performances through a downloadable end-of-game Comma-Separated Values (CSV) file that

students can submit. After the post-game survey, students are given the option to download a CSV file tracking all actions performed during the game (i.e. including which answers were answered corrently/incorrectly) and the pre- and post-game survey results. Instructors can parse this machine-readable format to draw useful information on the learning process experienced throughout the game by students.

Designing a session based on ‘Science Around the Board’

Designing a successful *Science Around the Board* session requires intentionality, ensuring that the activity, and questions, align with the learning objectives of the class/course.

Here is a framework to design your session and enhance the pedagogical impact.

Backward design: aligning with Learning Outcomes (LOs)

Before writing a single question, start with your Learning Outcomes . What exactly should students be able to do by the end of this session? Once your LOs are clearly defined, map them directly to the game's mechanics:

- **Foundational knowledge (properties):** Use standard property tiles to test basic recall and terminology. If an LO is "Define the function of 16S rRNA," this belongs on a property tile.
- **Synthesis and application (milestones):** The four corner Milestones represent mastery. Reserve your highest-order thinking questions for these exams. If an LO is "Analyze a pipeline output to troubleshoot denoising errors," this scenario may be better suited in a Milestone quiz.
- Use Lab Mishaps to teach practical realities or common pitfalls that do not fit neatly into a multiple-choice format (e.g., "You forgot to balance the centrifuge!").

Crafting the question file

When writing your TSV file, remember that the "incorrect" options are just as important as the correct one.

- **Write plausible distractors:** Do not use throwaway joke answers. Every incorrect option should represent a common student misconception. When a student chooses a distractor, it reveals a specific gap in their mental model.
- **Leverage the 'Explanation' column:** This is your most powerful teaching tool in the game. Do not just write "Incorrect." Use this space to immediately correct the specific

misconception tied to the distractors. Immediate, targeted feedback at the exact moment of failure is what transforms this from a quiz into a learning tool.

Structuring your time: The BOPPPS Model

To ensure the game serves an academic purpose, I highly recommend structuring your session around the **BOPPPS model** . For a standard 80-minute block, this translates to:

1. **Bridge-in (5 mins):** Hook the students. Explain *why* you are using a game today (e.g., to synthesize the last 4 weeks of material) and outline the victory conditions.
2. **Outcomes (2 mins):** Explicitly state the LOs on the board so students know what they are meant to be learning.
3. **Pre-assessment (10 mins):** Have students log in and complete the mandatory Pre-Game Survey. This establishes a metacognitive baseline for their confidence and current knowledge.
4. **Participatory Learning (45 mins):** This is the gameplay phase. Step back and become the Game Master. Let the competitive tension and "hard fun" drive the engagement.
5. **Post-assessment (10 mins):** Call "Time!" and have students complete the Post-Game Survey and export their data.
6. **Summary / Debrief (8 mins):** Do not skip this! Ask the room: *"Which side of the board was the hardest to defend?"* or *"What was the most surprising thing you learned from a mistake?"* Harvesting the play into conscious realization is critical.

Common challenges

1) Ensure students are not using their browsers on ‘Dark Mode’ as this prevent some of the questions from displaying correctly

Conclusion

Stepping away from the podium and handing control of the review session over to a board game can feel like a leap of faith. As instructors, we are often trained to deliver content efficiently, and the noisy, debate-filled environment of a gamified workshop is a stark contrast to a quiet lecture hall. However, as you will quickly discover during your first session with *Science Around the Board*, that noise is the sound of active, collaborative learning.

By leaning into playful pedagogy, we accomplish much more than simply reviewing material; we fundamentally change the students' relationship with the content. We replace the anxiety of high-stakes testing with the "hard fun" of strategic competition, loss aversion, and immediate feedback. The magic circle of the game board creates a space where failure is not only safe, but structurally necessary for growth.

Whether your students are defending the nuances of a microbiome data pipeline, troubleshooting laboratory equipment, or dominating a pop-culture trivia night, the underlying mechanism remains the same: they are engaged, they are thinking critically, and they are harvesting their play into conscious understanding.

Building a great TSV cartridge takes intentional design, and mastering the role of the game master takes practice. But the payoff, watching your students actively conceptualize complex material rather than passively memorizing it, is entirely worth the effort.

Welcome to the board, and have a great session!